

Vo Quoc Huy

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EXPERIENCE

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Ho Chi Minh City, Vietnam

Software Engineer (FastAPI, Azure, Terraform, PostgreSQL, Kubernetes, AI Agents)

May 2025 – Present

- Led the architecture and delivery of a distributed document synchronization platform, cutting processing latency from 15 minutes to 1 minute by replacing scheduled batch execution with a fault-tolerant, event-driven workflow.
- Owned the end-to-end Azure infrastructure lifecycle, automating UAT, staging, and production provisioning through Terraform/Terragrunt and approval-gated Azure DevOps pipelines; reduced environment setup from 1–2 days to minutes while improving deployment consistency and security through Managed Identity.
- Transformed production releases by introducing approval-based CI/CD and GitOps with ArgoCD, replacing manual deployments with declarative Kubernetes workflows and automated reconciliation.
- Modernized cloud infrastructure and reduced annual costs by approximately 50% by removing underutilized resources and optimizing storage lifecycle policies, spot-node usage, and compute tiers.
- Established end-to-end observability across distributed workflows using correlation IDs, business audit trails, application telemetry, and automated error alerting, improving production troubleshooting and incident diagnosis.
- Developed a proof-of-concept CDC pipeline (MySQL → Kafka → ClickHouse), reducing representative analytical queries from ~6 hours to ~30 seconds while evaluating OLAP adoption and production trade-offs.
- Built a document mapping system using Azure Document Intelligence and custom change-detection logic to extract structured data from high-volume PDF workloads, reducing manual processing effort.
- Enabled frontend applications and AI agents to invoke backend capabilities by integrating MCP and exposing standardized tool interfaces for service discovery and execution.

Employment Hero

Remote

Backend Software Engineer (Ruby on Rails, PostgreSQL, gRPC, CI/CD)

Apr 2024 – Apr 2025

- Designed a scalable search system using PostgreSQL (tsvector, trigram), providing a cost-effective alternative to expensive external search infrastructure.
- Identified and resolved a critical authentication vulnerability related to token misuse; collaborated with team leads to implement session-handling safeguards.
- Contributed to a 5x cost reduction in CI/CD by migrating all service pipelines from CircleCI to GitHub Actions.
- Optimized gRPC communication and complex database queries to resolve critical performance bottlenecks.
- Built a cross-device auto-save mechanism using Redis caching that directly improved platform retention metrics.

Athena Studio

Ho Chi Minh City, Vietnam

Software Engineer (Python, FastAPI, GCP, Airflow)

Nov 2022 – Mar 2024

- Developed high-throughput ingestion pipelines using FastAPI for automated revenue and cost data collection at scale.
- Implemented event-driven processing systems via GCP Pub/Sub, handling millions of daily events while cutting infrastructure costs by approximately 25%.

EDUCATION

Royal Melbourne Institute of Technology Vietnam

Ho Chi Minh City, Vietnam

Bachelor of Software Engineering (First Class Honours)

Sep 2019 – Sep 2023

SKILLS

Systems & Architecture: Distributed Systems, Event-Driven Architecture, Microservices, Scalable System Design.

Backend Development: Python (Django, FastAPI), Ruby on Rails, TypeScript, API Design (REST, gRPC, GraphQL).

Data & Streaming: PostgreSQL, MySQL, ClickHouse, Redis, Kafka, SQL Optimization, Data Pipelines, Airflow.

Cloud & Platform: Terraform, Terragrunt, Azure, GCP, Docker, Kubernetes, ArgoCD.

CI/CD: GitHub Actions, Azure DevOps.

AI Engineering: LLM Integration, RAG Architectures, AI Workflow Integration.

Languages: English (Full professional proficiency), Vietnamese (Native).